

Graph Adversarial Defense via Hilbert-Schmidt Independence Criterion against Influence Maximization Attacks

Yuxing Guo
School of Computer and Information
Technology
Shanxi University
Taiyuan, Shanxi, China
202312407003@email.sxu.edu.cn

Jianqing Liang^{*}
School of Computer and Information
Technology
Shanxi University
Taiyuan, Shanxi, China
liangjq@sxu.edu.cn

Kaixuan Yao
School of Computer and Information
Technology
Shanxi University
Taiyuan, Shanxi, China
yikx@sxu.edu.cn

Zhihao Guo
School of Economics and
Management
Shanxi University
Taiyuan, Shanxi, China
gzh081700@163.com

Jiye Liang
School of Computer and Information
Technology
Shanxi University
Taiyuan, Shanxi, China
ljy@sxu.edu.cn

Abstract

Graph Neural Networks (GNNs) demonstrate promising performance in data mining yet exhibit inherent vulnerabilities to adversarial attacks. Even imperceptible perturbations degrade model performance, seriously hindering the application of GNNs in reality. In recent years, adversarial defense methods based on model architecture have gained attention for their effectiveness. However, they exhibit limited effectiveness against emerging black-box influence maximization attacks (IMAs), which aim to maximize the spread of feature perturbations through a group of influential nodes. This may leave a potential risk in real-world applications. To address this issue, we propose a Graph Adversarial Defense method based on the Hilbert-Schmidt Independence Criterion (HSIC-GAD). Specifically, the proposed method leverages hidden representations to capture the dependence between preprocessed node features and label information. On this basis, we design a regularizer that simultaneously preserves the most relevant information for downstream tasks while filtering out adversarial perturbations from the input data. A simple theoretical analysis shows that the HSIC regularizer can reduce the sensitivity of the model to adversarial inputs. Additionally, it exhibits strong universality, consistently enhancing the adversarial robustness of diverse models. Extensive experiments on real-world datasets demonstrate that HSIC-GAD outperforms state-of-the-art defense methods against IMAs.

CCS Concepts

• Computing methodologies → Artificial intelligence.

^{*}Corresponding author



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Keywords

Graph neural networks, Influence maximization attacks, Adversarial defense

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1 Introduction

Graphs, characterized by node-edge relationships, can effectively represent real-world scenarios [1]. Graph data contains rich and complex information that is crucial for scientific research. Consequently, the effective and efficient extraction of such information receives significant attention and effort from the researchers. In recent years, GNNs [22], renowned for their message-passing mechanism, have achieved significant breakthroughs in graph data mining. Given their strong performance, GNNs exhibit great potential for a wide range of applications, including recommendation systems [38] and autonomous driving [16], among others [9, 29]. However, the vulnerability [12] of GNNs raises concerns about their performance and reliability, given that even imperceptible perturbations can significantly affect their predictions.

In response, researchers have developed various adversarial defense strategies, such as adversarial training [7, 10, 35], attack detection [28, 33, 39] and model architectures [3, 17, 20, 25, 40]. Among these, model architecture-based [34] adversarial defense methods preserve inherent properties of graphs—such as low-rankness, sparsity, and smoothness—during training [20], and typically modify the message aggregation mechanisms [17] or design novel loss functions [25, 40] to enhance adversarial robustness. This class of methods has attracted significant attention in recent research due to its generality and effectiveness. Therefore, this work primarily focuses on model architecture-based adversarial defense methods.

Although adversarial defense methods built upon model architecture have achieved strong performance against various types

of attacks, extensive experimental results reveal that they exhibit limited effectiveness against IMAs [26, 27, 32]. IMAs represent a novel class of black-box adversarial strategies that formulate the attack as an influence maximization problem [21], selecting a set of influential nodes to maximize the propagation of feature perturbations across the graph. As shown in Figure 1, under the IMA, as the attack strength increases, the predictive performance of the model continuously declines, yet the dependence between node representations and labels gradually strengthens. This indicates that the IMAs induce the model to capture spurious dependencies, thereby leading to a substantial drop in prediction accuracy. This poses security risks and may undermine the reliability and safety of GNNs in real-world applications. Motivated by this observation, we explore adversarial robustness by guiding the model toward learning meaningful and correct dependencies. To this end, we face the following challenges: how to purify node features when the perturbation knowledge, including perturbation methods, strengths, and target nodes, is unknown; how to effectively measure dependencies between representations given the inconsistency in representation dimensions across GNN layers; and how to leverage dependence to learn adversarial robust node representations while removing node noise. To address these challenges, we propose a novel Graph

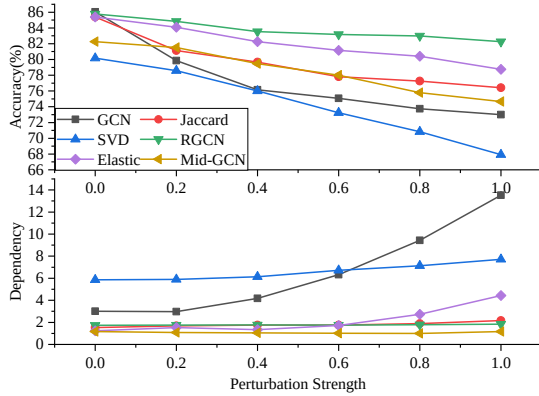


Figure 1: The variation of model accuracy and the dependency between node representations and labels with the strength of IMAs.

Adversarial Defense method based on the HSIC (HSIC-GAD). It primarily comprises a local neighborhood smoothing (LNS) module and an HSIC regularization module, the latter designed to hinder the propagation of adversarial perturbations while preserving information relevant to downstream tasks. Motivated by the limitation of perturbation budgets, we propose a parameter-free approach that leverages local neighborhood information to enhance the adversarial robustness of individual nodes. The contributions are summarized as follows:

- We propose a novel adversarially robust GNN method to address the lack of effective defense against emerging black-box IMAs.
- We leverage HSIC to retain essential information for output prediction while minimizing redundancy, building on local neighborhood smoothing. A simple yet effective theoretical

proof demonstrates that HSIC regularization enables models to be less sensitive to perturbations.

- Extensive experiments on real-world datasets demonstrate that HSIC-GAD provides state-of-the-art robustness against IMAs and generalizes well to other adversarial scenarios.

2 Related Work

2.1 Graph Adversarial Attacks

Numerous adversarial attack works [5, 6, 26, 27, 42] aim to expose the vulnerabilities of GNNs. These attacks can be categorized in various ways based on different classification criteria [19, 28]. For example, adversarial attacks can be classified as poisoning attacks [42], which occur before training, or evasion attacks [4, 7], which occur after training. They may also be targeted attacks [7, 41], which aim to misclassify specific nodes, or untargeted attacks [42], designed to degrade the overall performance of models. Furthermore, attacks can operate in a white-box setting [5, 6, 24], where all model information is accessible, or in a black-box setting [7, 26, 27], where the model details remain unknown. Feature attacks directly manipulate node features, while topology attacks alter node representations indirectly by modifying graph structure. In addition, hybrid attacks combine both feature and topology perturbations. To better reflect real-world scenarios, researchers have increasingly focused on black-box untargeted evasion attacks.

Recently, IMAs, a type of black-box untargeted evasion attack, have attracted considerable attention. Attackers leverage a carefully selected set of nodes to propagate feature perturbations across the entire graph, thereby affecting the majority of nodes and substantially reducing the prediction accuracy of the model. Ma et al. [26, 27] frame the adversarial attacks as an influence maximization problem, introducing methods such as RWCS, GC-RWCS, InfMax-Unif, and InfMax-Norm. Building on this concept, Wen et al. [32] propose VoteAttack, a voting mechanism to identify key nodes for influence maximization.

Although this type of attack is still in its early stages, it has been shown to be highly effective in degrading model performance. Existing defense strategies largely overlook its potential threat. This oversight raises serious concerns about the real-world robustness of GNNs, particularly in safety-critical applications such as finance, healthcare.

2.2 Graph Adversarial Defense

To enhance the adversarial robustness of GNNs and thereby facilitate their widespread deployment in real-world applications, researchers have proposed various defense strategies [19, 28, 34] based on the vulnerabilities exposed by adversarial attacks. Dai et al. [7] simulates adversarial attacks by randomly adding or removing edges during model training to enhance adversarial robustness, which inspires adversarial training [10, 35] for graphs. However, this strategy requires retraining the model, making it computationally expensive. Moreover, its effectiveness is constrained by the types of adversarial examples used during training. Subsequent researchers have proposed adversarial defense strategies for attack detection by leveraging the rich and well-established prior knowledge in graph data research, including attribute similarity between connected nodes [33], degree distribution similarity [39],

among others [28]. These types of methods are model-agnostic, but detection performance depends on the threshold setting. To overcome the above limitations, researchers have shifted their focus to adversarial defense methods based on model architecture. These methods modify the message aggregation mechanisms [25, 40] or introduce new loss functions [3, 17, 20].

The third category of methods [34] offers strong interpretability, broad defense coverage, and high effectiveness. Therefore, in this work, we primarily focus on this category. Different from prior work, we consider defenses against IMAs. We leverage HSIC to capture similarity across different representational spaces and different feature dimensions, and introduce an HSIC-based regularization mechanism to promote adversarially robust node representations.

2.3 Hilbert-Schmidt Independence Criterion

The Hilbert-Schmidt Independence Criterion (HSIC) [14] is defined as the Hilbert-Schmidt norm of the cross-covariance operator between distributions in reproducing kernel Hilbert spaces (RKHSs). He et al. [15] utilized HSIC for learning effective representations. Greenfeld et al. [13] utilized HSIC as a loss function for robust regression and classification to mitigate the challenge of unsupervised covariate shift. Building upon this, Wang et al. [31] introduced HBaR, which investigates the use of HSIC as a regularizer to enhance the adversarial robustness of visual models. Subsequently, Jian et al. [18] and Zhang et al. [37] extended the application of HSIC regularization to adversarial pruning and automatic modulation classification tasks, with the goal of improving robustness in these contexts.

Different from previous work, this paper is the first to apply HSIC to the study of adversarial robustness in GNNs, providing an effective method for measuring dependencies between variables in different spaces and combining it with local neighborhood smoothing to mitigate the propagation of perturbations in the message-passing mechanism.

3 Preliminaries

3.1 Notations

We represent a graph as $G = (A, X)$, where $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix, where $A_{ij} = 1$ indicates an edge between v_i and v_j , and $X \in \mathbb{R}^{N \times D}$ denotes the node feature matrix with D -dimensional features per node. The node set is $V = \{v_1, v_2, \dots, v_N\}$, where N is the number of nodes. In node classification, V_L and V_U denote the labeled and unlabeled nodes, respectively, and K is the number of classes.

3.2 Graph Adversarial Robustness

The goal of the node classification for GNNs is to use a model $f_\theta(\cdot, \cdot)$ trained on labeled nodes V_L to efficiently predict the labels of unlabeled nodes V_U in the graph. The objective function can be formulated as:

$$\min_{\theta} \mathcal{L}_{train}(f_\theta(G)) = \sum_{i \in V_L} \mathcal{L}(f_\theta(A, X)_i, Y_i), \quad (1)$$

where θ is the parameter of the model, $f_\theta(\cdot, \cdot)_i$ is the prediction of node v_i , Y_i is the true label of node v_i and $\mathcal{L}(\cdot, \cdot) : \mathbb{R}^K \times \mathbb{R}^K \rightarrow \mathbb{R}$ is to measure the difference between prediction and true label such as

cross-entropy loss. To make the perturbations more imperceptible, attackers often give a fixed perturbation budget Δ , i.e.,

$$\hat{G} = (\hat{A}, \hat{X}) \text{ s.t. } \|\hat{A} - A\|_0 + \|\hat{X} - X\|_0 \leq \Delta, \quad (2)$$

where $\|\cdot\|_0$ is the L0 norm, $\hat{A}$, $\hat{X}$, $\hat{G}$ denote the perturbed adjacency matrix, node attribute matrix, and graph, respectively. We aim to find a model $f_\theta(\cdot, \cdot)$ that has high prediction accuracy but is also adversarially robust: the model should maintain high prediction accuracy even when input samples are subjected to perturbations within a fixed budget.

3.3 HSIC

The HSIC [13, 14] is a statistical method used to measure the independence between two random variables X and Z , which reside in the metric space $\mathcal{X}$ and $\mathcal{Z}$, respectively. $\mathcal{F}$ and $\mathcal{G}$ are two separable RKHSs over $\mathcal{X}$ and $\mathcal{Z}$, respectively, and are capable of assessing the nonlinear dependencies between them. HSIC (X, Z) is defined as:

$$\text{HSIC}(X, Z; \mathcal{F}, \mathcal{G}) \equiv \|C_{XZ}\|_{HS}^2. \quad (3)$$

Using kernel functions, HSIC (X, Z) can be estimated in an unbiased manner from finite samples, it can be formulated as:

$$\text{HSIC}(X, Z) = \frac{1}{(n-1)^2} \text{tr}(K_X J K_Z J), \quad (4)$$

where n is the number of samples drawn from the joint distribution on $\mathcal{X} \times \mathcal{Z}$, $J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T$, I is an identity matrix, $\mathbf{1}$ is an all 1's matrix. The kernel matrix $K_X \in \mathbb{R}^{n \times n}$ consists of entries k_{x_i, x_j} , where each entry is computed using the Gaussian kernel $k_{x_i, x_j} = \exp(-\frac{\|x_i - x_j\|_2^2}{2\sigma^2})$, where x_i, x_j denote node features, the computation of K_Z is the same as that of K_X , and σ is a constant.

4 Methodology

This section introduces an adversarial robust GNN method that leverages local neighborhood smoothing and HSIC regularization to defend against IMAs. In Section 4.1, we present the local neighborhood smoothing. In Section 4.2, we introduce the HSIC to measure the correlations between hidden representations and both node features and labels. Based on this, we design a novel loss function to enhance the adversarial robustness of the model. A straightforward robustness guarantee based on HSIC is provided in Section 4.3. Finally, we outline an objective function and analyze the time complexity in Section 4.4.

The success of IMAs largely relies on the message-passing mechanism of GNNs to maximally propagate feature perturbations throughout the graph. Moreover, since perturbation budgets are limited in adversarial attacks, most nodes in the graph remain unperturbed. A logical defense strategy is to purify node features and hinder the propagation of residual perturbation. Figure 2 gives a detailed illustration of our framework.

4.1 Local Neighborhood Smoothing

Building on insights from related work [20] and the above understanding, we recognize the importance of clean node features in training GNNs. Therefore, without prior knowledge of perturbations, we can utilize the neighborhood information around the

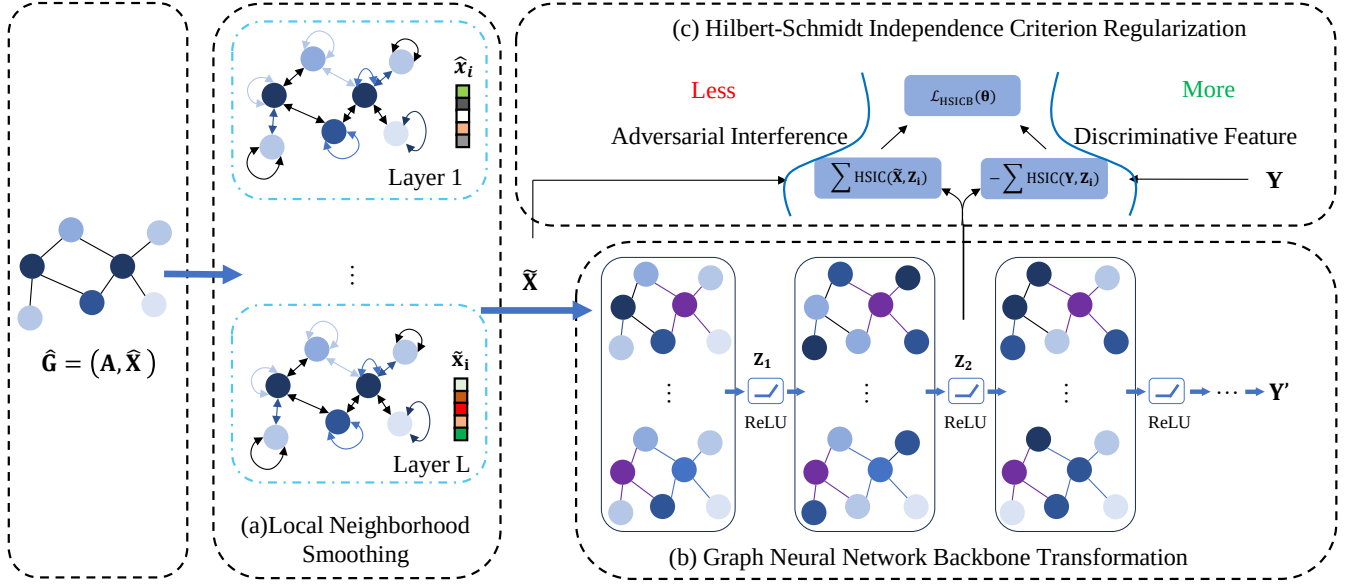


Figure 2: The overall framework of our proposed HSIC-GAD: (a) Local Neighborhood Smoothing, which refines node features by leveraging multi-order information from neighboring nodes; (b) Transformation, which aggregates information via various backbone models; (c) HSIC-based Regularization, which restricts the spread of residual perturbations. Finally, the model learns a minimal sufficient representation by the optimization of $\mathcal{L}$.

nodes to enhance the node features. Even if the perturbation information is transmitted to the central node, similar to a voting mechanism, the number of undisturbed nodes around the central node outweighs the perturbed nodes, thus the central node is less affected by the perturbation. If the central node itself is a perturbed node, then local neighborhood smoothing can utilize the surrounding nodes to dilute the adversarial perturbation. Unlike the message passing mechanism, this scheme is computationally efficient and requires no additional parameters to learn, and effectively leverages different-order neighbor information to purify node features. The formal representation is as follows:

$$\begin{aligned} \mathbf{X}^{(0)} &= \hat{\mathbf{X}}, \\ \mathbf{X}^{(l+1)} &= \alpha \mathbf{X}^{(0)} + (1 - \alpha) \tilde{\mathbf{A}} \mathbf{X}^{(l)}, \\ \tilde{\mathbf{X}} &= \mathbf{X}^{(L)}, \end{aligned} \quad (5)$$

where $\hat{\mathbf{X}}$ is the input, $\alpha \in (0, 1]$ is the teleport probability, $\tilde{\mathbf{A}}$ is the symmetrically normalized adjacency matrix with self-loops, l is the number of iteration steps.

After local neighborhood smoothing, the transformation in Gaussian GNNs [2, 40] maps $\tilde{\mathbf{X}}$ to the predicted labels $\mathbf{Y}'$. As shown in Appendix A.2.1, the proposed method is equally effective across various backbone models. We take Gaussian GNNs [40] as an example to demonstrate the effectiveness of the HSIC-GAD. The loss function of the Gaussian GNN model is denoted as $\mathcal{L}_{Gau}$.

4.2 HSIC Regularization for Adversarial Robustness

The literature [14] indicates that HSIC possesses several desirable properties. Specifically, HSIC can measure the dependence between

two random variables in arbitrary spaces. Moreover, HSIC leverages the entire spectrum of the cross-covariance operator to assess dependence. Compared to other dependence evaluation methods, HSIC demonstrates greater robustness in the presence of outliers or noise, making its dependency assessments more reliable. These properties make HSIC suitable for capturing the dependence relationships in complex data.

Given H layers Gaussian GNNs $f_{\theta}(\mathbf{A}, \tilde{\mathbf{X}}) : \mathbb{R}^{N \times N} \times \mathbb{R}^{N \times D} \rightarrow \mathbb{R}^{N \times K}$ parameterized by θ . We add the HSIC regularization in every layer of this model, which is represented as follows:

$$\mathcal{L}_{HSIC} = \lambda_x \sum_i \text{HSIC}(\tilde{\mathbf{X}}, \mathbf{Z}_i) - \lambda_y \sum_i \text{HSIC}(\mathbf{Y}, \mathbf{Z}_i), \quad (6)$$

where $\mathbf{Z}_i \in \mathbb{R}^{N \times D_{Z_i}}$ is the output of i -th layer, λ_x, λ_y denote the hyperparameters used to balance the intermediate representations between relevant information to the prediction and redundant information from the preprocessed input $\tilde{\mathbf{X}}$. A larger HSIC value indicates a strong correlation between the two random variables, while a smaller value suggests a weak or no correlation between them. Leveraging this property of HSIC, minimizing $\text{HSIC}(\tilde{\mathbf{X}}, \mathbf{Z}_i)$ reduces the correlation between the learned representation at the i -th layer and the preprocessed input, thereby filtering out perturbations added to the input data. Similarly, maximizing $\text{HSIC}(\mathbf{Y}, \mathbf{Z}_i)$ encourages the retention of information most relevant to downstream tasks. The HSIC value can be computed using Eq. (4). Balancing $\text{HSIC}(\tilde{\mathbf{X}}, \mathbf{Z}_i)$ and $\text{HSIC}(\mathbf{Y}, \mathbf{Z}_i)$ is essential for achieving adversarial robustness.

4.3 HSIC-GAD Robustness Guarantees

Robust learning with HSIC [31] facilitates the acquisition of minimal sufficient representations. Building on this, we extend the application of HSIC to GNNs to improve their robustness. We provide a formal justification that HSIC reduces the impact of input perturbations on model performance, thereby enhancing the robustness of HSIC-GAD.

Lemma 1. [13] Suppose that $\mathcal{F}, \mathcal{G}$ are two separable RKHSs on $\mathcal{X}, \mathcal{Z}$ induced by k_X and k_Z , respectively. Suppose X and Z are random variables residing in metric spaces $\mathcal{X}, \mathcal{Z}$, respectively. Then, the following inequality holds:

$$\text{HSIC}(X, Z; \mathcal{F}, \mathcal{G}) \geq \sup_{\zeta \in \mathcal{F}, \eta \in \mathcal{G}} \text{Cov}[\zeta(X), \eta(Z)]. \quad (7)$$

Lemma 2. [14] Let $\mathcal{F}, \mathcal{G}$ be separable RKHSs over $\mathcal{X}, \mathcal{Z}$ respectively, s.t. $\|\zeta\|_{\mathcal{F}} \leq M_{\mathcal{F}}$ for all $\zeta \in \mathcal{F}$ and $\|\eta\|_{\mathcal{G}} \leq M_{\mathcal{G}}$ for all $\eta \in \mathcal{G}$. Then the follow holds:

$$\begin{aligned} & \sup_{\zeta \in \mathcal{F}, \eta \in \mathcal{G}} \text{Cov}[\zeta(X), \eta(Z)] \\ &= M_{\mathcal{F}} M_{\mathcal{G}} \sup_{\zeta \in \tilde{\mathcal{F}}, \eta \in \tilde{\mathcal{G}}} \text{Cov}[\zeta(X), \eta(Z)], \end{aligned} \quad (8)$$

where $\tilde{\mathcal{F}}$ and $\tilde{\mathcal{G}}$ be the restriction of $\mathcal{F}$ and $\mathcal{G}$ to functions in the unit ball of the respective RKHS through rescaling.

Theorem 1. Assume $\tilde{X} \in \mathcal{X}, Z \in \mathcal{Z}$ are random variables residing in metric spaces $\mathcal{X}, \mathcal{Z}$. Denote by $\mathcal{F}$ and $\mathcal{G}$ RKHSs of functions from $\mathcal{X}$ and $\mathcal{Z}$, s.t. $\|\zeta\|_{\mathcal{F}} \leq M_{\mathcal{F}}, \|\eta\|_{\mathcal{G}} \leq M_{\mathcal{G}}$, we have

$$\begin{aligned} & \text{HSIC}(\tilde{X}, Z; \tilde{\mathcal{F}}, \tilde{\mathcal{G}}) \\ & \geq \sup_{\zeta \in \tilde{\mathcal{F}}, \eta \in \tilde{\mathcal{G}}} \text{Cov}(\zeta(\tilde{X}), \eta(Z)) \\ &= \frac{1}{M_{\mathcal{F}} M_{\mathcal{G}_i}} \lim_{n \rightarrow \infty} \text{Cov}(\zeta_n(\tilde{X}), Z_i) \\ &= \frac{1}{M_{\mathcal{F}} M_{\mathcal{G}_i}} \text{Cov}(Z_i, Z_i) \\ &= \frac{1}{M_{\mathcal{F}} M_{\mathcal{G}_i}} \text{Var}(Z_i), \end{aligned} \quad (9)$$

where $\tilde{X}$ is the output after LNS, assume that there exists a sequence of functions $\{\zeta_n\}_{n=1}^{\infty}$ which converges uniformly to Z_i , $\mathcal{G}_i$ ($i = 1, \dots, h$) denote the RKHSs to which the representation of the i -th layer of GNNs belongs, $\tilde{\mathcal{F}}$ and $\tilde{\mathcal{G}}_i$ are the restrictions of $\mathcal{F}$ and $\mathcal{G}_i$ to functions in the unit ball of the respective RKHSs through rescaling. Additionally, for all $Z_i \in \mathcal{G}_i$, we have $\|Z_i\|_{\mathcal{G}_i} \leq M_{\mathcal{G}_i}$.

The above justification indicates that applying HSIC to any intermediate layer will control the variation of the output. Based on this observation, we utilize HSIC(Y, Z_i) to extract useful information for downstream tasks.

4.4 Objective Function of the HSIC-GAD and Complexity Analysis

By combining the above two loss functions with the weighting coefficients λ_x, λ_y the overall objective of the HSIC-GAD is

$$\mathcal{L} = \mathcal{L}_{\text{Gau}} + \lambda_x \sum_i \text{HSIC}(\tilde{X}, Z_i) - \lambda_y \sum_i \text{HSIC}(Y, Z_i). \quad (10)$$

Algorithm 1: HSIC-GAD: Hilbert-Schmidt Independence Criterion for Graph Adversarial Defense.

Input: Noisy graph $\hat{G} = (A, \hat{X})$, kernel functions K_X, K_Z, K_Y , model f_{θ} , learning rate β , hyperparameters $\alpha, l, \lambda_x, \lambda_y$.

Output: Model parameters θ and hidden representations $\{Z_i\}_{i=1}^H$

Initialize parameters θ ;

Local neighborhood smoothing via Eq. (5);

while not converged **do**

 Forward pass: compute $f_{\theta}(\cdot, \cdot), Z_i$;

 Compute kernel matrices K_X, K_Z, K_Y ;

 Compute loss $\mathcal{L}$ via Eq. (10);

 Update $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}$;

return θ

The overall algorithm is described in Algorithm 1. This framework is trained end-to-end and can be optimized using stochastic gradient descent algorithms. The time complexity of our framework is $O(|E| \sum_{h=0}^H D_{z_h} + N \sum_{h=1}^H D_{z_h} D_{z_{h-1}} + N^2 \sum_{h=1}^H D_{z_h})$ where $N = |V|$ is the number of nodes, $|E|$ is the number of edges, D_{z_h} is the dimension of the h -th hidden layer. To reduce the computational complexity of HSIC, this paper employs a mini-batch and random neighbor sampling strategy. Each batch contains m samples, and the computational complexity per batch for a single layer is $O(m^2 \sum_{h=1}^H D_{z_h})$. Therefore, the overall computational complexity of HSIC is $O(\lceil \frac{N}{m} \rceil m^2 \sum_{h=1}^H D_{z_h})$. The computation supports high parallelization, and with the help of the above strategy, the computational cost of the model is reduced.

5 Experiments

In this section, we evaluate the effectiveness of the HSIC-GAD against different graph adversarial attack strategies on various real-world datasets to answer the subsequent research questions:

- **RQ1:** How does HSIC-GAD perform in comparison to the state-of-the-art robust models under IMAs?
- **RQ2:** How does each component of the HSIC-GAD contribute to adversarial robustness?
- **RQ3:** How do key parameters influence its robustness performance?

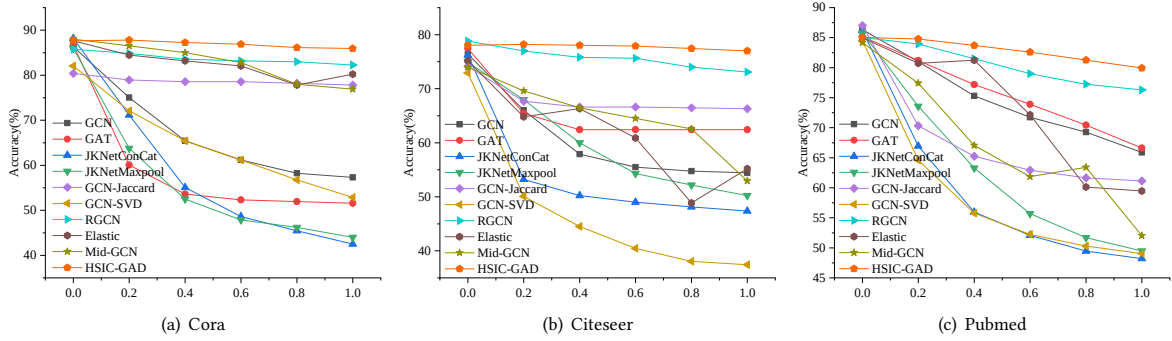
5.1 Experimental Settings

5.1.1 Datasets. We validate the effectiveness of the HSIC-GAD on widely used publicly available benchmark datasets, i.e., Cora, Cora_ML, Citeseer, Pubmed, Polblogs, ACM, and BlogCatalog. The basic statistics of these datasets are provided in the Appendix A.1.1. To maintain consistency, the data split ratio is kept the same across all datasets. The source code of this paper has been made publicly available at <https://github.com/YuxingGuo/HSIC-GAD>.

5.1.2 Baseline and Adversarial Attack Methods. To evaluate the effectiveness of the HSIC-GAD, we compare it with three typical GNNs and five robust GNNs under IMAs and other attack strategies,

Table 1: Node classification performance (accuracy (%)) under clean and adversarial settings

Dataset	Type	Model Method	GCN	GAT	JKNet	Jaccard	SVD	RGCN	Elastic	Mid-GCN	Ours
Cora	Clean	None	83.03	85.73	82.01	80.41	80.57	<u>85.77</u>	84.82	82.26	87.43
	Centrality	Degree	85.03	<u>85.40</u>	71.16	85.39	54.60	82.43	85.03	81.89	87.43
		PageRank	73.57	<u>83.55</u>	49.72	73.38	65.62	82.43	78.93	72.46	86.13
		Between	75.60	<u>82.81</u>	56.19	76.16	66.36	81.70	78.93	73.94	84.84
	IMAs	Random	73.57	81.89	76.89	76.34	64.51	<u>82.07</u>	77.26	72.64	84.84
		GC-RWCS	74.49	<u>82.62</u>	42.51	78.93	67.51	<u>82.07</u>	79.30	72.27	85.76
		RWCS	73.57	<u>83.36</u>	49.72	73.94	65.67	82.25	79.30	73.38	85.95
		InfMax-Unif	73.01	81.89	43.25	76.71	64.88	<u>82.25</u>	76.71	72.09	85.95
		InfMax-Norm	73.94	<u>82.81</u>	42.33	77.26	64.01	81.88	78.56	73.01	85.39
		VoteAttack	73.01	82.25	42.51	72.64	65.71	<u>82.99</u>	78.74	74.68	85.95
	Gradient	IG-FGSM	37.50	<u>82.50</u>	<u>82.50</u>	<u>82.50</u>	72.50	87.50	<u>82.50</u>	80.00	87.50
		Nettack	47.50	60.00	70.00	55.00	62.50	62.50	60.00	<u>77.50</u>	82.50
Citeseer	Clean	None	74.89	73.08	73.21	74.14	70.00	78.80	74.44	71.28	78.50
	Centrality	Degree	68.27	<u>74.59</u>	60.60	68.12	54.64	72.78	72.33	65.11	78.65
		PageRank	54.43	69.62	47.82	58.05	57.09	<u>71.27</u>	58.35	60.90	78.50
		Between	60.30	72.33	49.77	60.15	56.77	<u>74.13</u>	61.35	62.41	77.74
	IMAs	Random	68.42	73.08	60.30	68.12	57.06	<u>76.54</u>	69.47	68.42	77.74
		GC-RWCS	52.78	70.83	45.71	54.59	59.55	<u>74.28</u>	54.89	58.64	78.35
		RWCS	54.89	69.62	47.82	58.80	55.89	<u>71.42</u>	58.50	61.20	78.50
		InfMax-Unif	54.59	71.58	47.97	54.29	59.06	<u>73.38</u>	55.64	59.55	78.50
		InfMax-Norm	53.38	71.13	45.71	53.68	57.28	<u>73.83</u>	55.64	58.50	78.35
		VoteAttack	50.23	70.38	47.37	54.88	56.91	<u>73.08</u>	55.64	58.65	78.35
	Gradient	IG-FGSM	20.00	87.50	87.50	87.50	87.50	80.00	<u>85.00</u>	<u>85.00</u>	87.50
		Nettack	32.50	62.50	67.50	65.00	65.00	72.50	72.50	<u>75.00</u>	77.50
Pubmed	Clean	None	87.95	85.20	86.29	87.07	84.07	85.04	86.25	84.39	85.65
	Centrality	Degree	73.78	<u>80.92</u>	54.53	62.97	44.23	80.52	58.94	60.93	82.68
		PageRank	69.34	67.93	48.82	68.68	59.26	<u>77.43</u>	56.89	68.04	79.81
		Between	66.85	70.55	49.30	67.39	58.33	<u>78.21</u>	57.27	70.62	80.65
	IMAs	Random	78.32	74.25	62.95	81.41	77.25	<u>82.42</u>	80.17	79.23	83.74
		GC-RWCS	64.82	70.88	47.25	67.16	57.04	<u>79.12</u>	58.66	71.59	81.59
		RWCS	67.61	67.38	48.26	66.37	57.61	<u>77.47</u>	56.30	68.36	80.07
		InfMax-Unif	64.95	71.87	46.49	68.25	60.76	<u>78.64</u>	58.74	72.14	81.18
		InfMax-Norm	62.59	71.18	46.01	66.14	58.34	<u>78.87</u>	57.80	71.78	81.31
		VoteAttack	63.65	65.91	48.21	62.95	53.28	<u>76.29</u>	56.81	69.74	79.93
	Gradient	IG-FGSM	72.50	80.00	77.50	77.50	<u>85.00</u>	<u>85.00</u>	<u>85.00</u>	82.50	90.00
		Nettack	70.00	55.00	62.50	70.00	67.50	67.50	70.00	<u>72.50</u>	77.50

**Figure 3: Results under different perturbation strengths on the VoteAttack.**

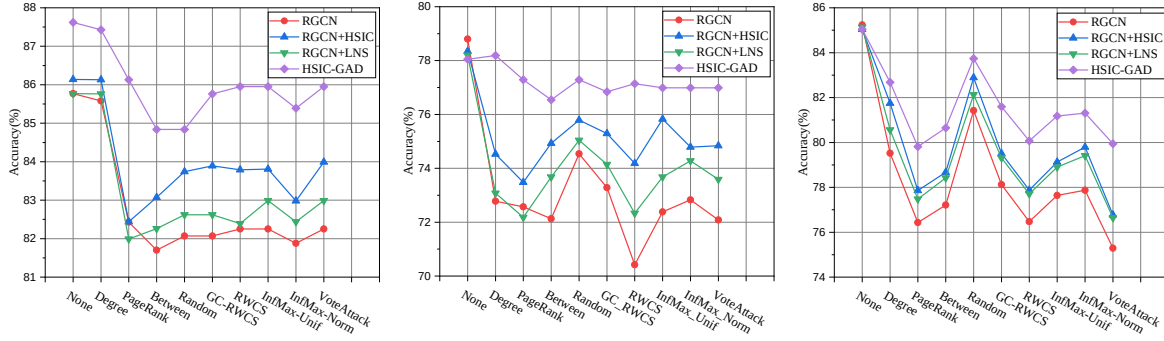


Figure 4: Ablation studies on Cora, Citeseer, and Pubmed.

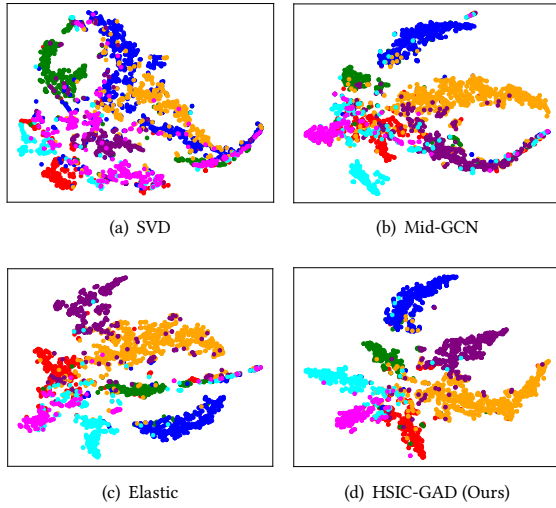


Figure 5: T-SNE visualizations of embeddings obtained from various methods on the Cora dataset.

using the GNNs library PyG[11] and the PyTorch adversarial library DeepRobust[23]. Details of the benchmark models and attack methods are provided in Appendix A.1.2 and Appendix A.1.3, respectively. The evaluation metrics are presented in Appendix A.1.4.

5.1.3 Parameter Settings. To ensure a fair comparison, we set the perturbation parameters to follow the settings of adversarial attacks [20, 26, 27, 32]. For IMAs, we randomly choose 60% of nodes for training, 20% for validation, and the remaining 20% for testing. For topology attacks, we use 10% of nodes for training, 10% for validation, and the remaining 80% for testing. In IMAs [26, 27, 32], the number of perturbation nodes is fixed at 1% of the total number of nodes in the graph. For each perturbation node, 2% of its feature dimensions are modified, and the perturbation strength is set to 1.

5.2 Evaluation on Adversarial Robustness (RQ1)

To answer RQ1, we evaluate the adversarial robustness of the HSIC-GAD using the attack methods mentioned above. The experiments

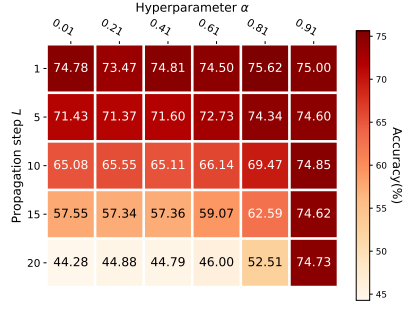
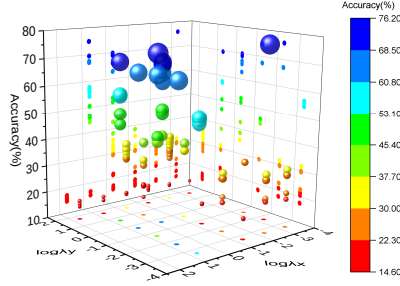
reveal that the proposed framework achieves the best performance against IMAs and demonstrates an advantage in defending against other types of adversarial attacks. A detailed analysis of the experimental results is provided below.

5.2.1 Adversarial Robustness under IMAs and Hybrid Attacks. We evaluate the adversarial robustness of HSIC-GAD against three types of attacks: Centrality, IMAs, and Gradient. Here, Centrality refers to a type of node selection strategy, such as Degree [27], PageRank, and Betweenness. Gradient includes Hybrid attacks. The experimental results are shown in Table 1, where the best and runner-up adversarial robustness performances are highlighted in bold and underlined, respectively. The clean metric denotes the performance of various models on clean data. HSIC-GAD performs on par with current state-of-the-art methods on clean datasets, notably achieving the best results on the Cora dataset.

In terms of defense, HSIC-GAD achieves the best performance against emerging IMAs and also demonstrates strong robustness against other feature attacks, including hybrid attacks. Specifically, the framework achieves an average 15.36% improvement in robustness against IMAs, demonstrating superior robustness against this emerging threat. In addition, it achieves an average robustness improvement of 14.79% against other types of feature attacks. These results demonstrate that HSIC-GAD effectively mitigates the impact of adversarial perturbations, thereby enhancing the overall robustness of the model.

The visualization in Figure 5 further supports these findings. To the best of our knowledge, VoteAttack is the most recent IMA, we employ T-SNE to illustrate the robustness of the proposed framework under its influence. The results show that the boundaries between different classes of nodes in the HSIC-GAD view are more distinct, revealing its robustness.

5.2.2 Adversarial Robustness Under Different Perturbation Strength. To demonstrate that HSIC-GAD is resistant to different levels of perturbation, we vary the perturbation strength across $\{0.2, 0.4, 0.6, 0.8, 1\}$ for IMAs and compare its robustness with that of benchmark models under the same settings. Figure 3 compares the robustness of different models under varying feature perturbation strengths, with VoteAttack applied as the attack method. As expected, the performance of all baseline models declines as the perturbation strength increases. Although the performance of HSIC-GAD also degrades,

(a) Sensitivity to α and L (b) Sensitivity to λ_x and λ_y **Figure 6: Parameter analysis on the Citeseer dataset.**

it remains considerably more stable and consistently outperforms all other models. Similar results are observed under extreme perturbation budgets, as detailed in Appendix A.2.3. HSIC-GAD also outperforms existing models in defending against topological attacks under varying perturbation strengths. The experimental results and analysis are provided in Appendix A.2.2. We attribute the robustness of the HSIC-GAD to local neighborhood smoothing and HSIC regularization. The former leverages different-order information to enhance the robustness of nodes themselves, and the latter reduces the propagation of residual perturbations, thereby mitigating the impact of adversarial attacks on model performance.

5.3 Ablation Study (RQ2)

Each component of the HSIC-GAD contributes to enhancing model robustness, and their combination further improves it. To verify this, we conduct ablation experiments on three datasets to answer RQ2. Four settings are tested: (1) Only the Gaussian GNN, abbreviated as RGCN; (2) with local neighborhood smoothing; (3) with HSIC regularization; and (4) with both modules.

As shown in Figure 4, removing either component significantly degrades robustness, while their combination yields superior performance. The two modules are complementary to each other. The local neighborhood smoothing module aggregates neighbor information in an equal proportion manner, which can dilute adversarial perturbations and enhance node attributes. On this basis, the HSIC regularization module reduces the propagation of perturbation information while retaining information relevant to downstream tasks.

The effectiveness of the proposed framework is independent of the choice of backbone models. Detailed results are provided in Appendix A.2.1.

5.4 Parameter Analysis (RQ3)

In this subsection, we explore the sensitivity of key hyperparameters in HSIC-GAD to address RQ3. Specifically, α controls the proportion of purified node attributes retained at each propagation step, L controls the number of propagation steps, λ_x regulates the extent of redundant information removal, and λ_y manages the dependency between the representations and their corresponding labels. Specifically, we vary the hyperparameters as follows: α in $\{0.01, 0.21, 0.41, 0.61, 0.81, 0.91\}$, L in $\{1, 5, 10, 15, 20\}$, λ_x and λ_y in $\{5 \times 10^1, 5, 5 \times 10^{-1}, 5 \times 10^{-2}, 5 \times 10^{-3}, 5 \times 10^{-4}\}$. Sensitivity analysis is performed on the Citeseer dataset, where accuracy is averaged across clean data, Metattack-perturbed data, and VoteAttack-perturbed data. Similar trends are observed on other datasets. The results shown in Figure 6 suggest the following:

(1) Figure 6(a) illustrates that appropriately leveraging neighbor information, while preserving the original node attributes, can reduce the impact of perturbations and enhance model robustness. Incorporating high-order neighbors yields better performance only when a relatively high value of α is maintained, which preserves more of the original node attributes during propagation.

(2) The parameters λ_x and λ_y are set to balance $\text{HSIC}(\tilde{X}, Z_i)$ and $\text{HSIC}(Y, Z_i)$ in the HSIC regularization. Balancing these two parameters has a positive effect on the model's adversarial robustness. Figure 6(b) shows that when other parameters remain unchanged, the model's adversarial robustness increases with the increase of λ_x within a certain range; conversely, the model's adversarial robustness decreases with the increase of λ_y . Moreover, when $\lambda_x < 1$ and $\lambda_y > 1$, the model's adversarial robustness drops significantly. This also indicates that appropriately reducing the value of $\text{HSIC}(\tilde{X}, Z_i)$ helps enhance adversarial robustness, while appropriately increasing the value of $\text{HSIC}(Y, Z_i)$ helps retain label-related information in the representation.

6 Conclusion

In this work, we propose a novel defense method, addressing the lack of effective defenses against emerging IMAs. It effectively leverages the HSIC to evaluate the similarity between node representations and node attributes or labels across different spaces and different dimensions. To prevent the propagation of perturbations, we design an HSIC-based regularizer that guides the model to learn minimal sufficient representations. Additionally, we incorporate a simple local neighbor smoothing module to preprocess the input data. Evaluations and experimental results demonstrate HSIC-GAD's state-of-the-art performance in defending against IMAs. Moreover, we find that the proposed method is effective against both topology and hybrid attacks. Future research directions include improving the efficiency of HSIC-GAD and exploring its effectiveness against graph injection attacks.

7 Acknowledge

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A Experimental Details and Additional Results

A.1 Experimental Details

A.1.1 Dataset and Implementation Details. Cora, Cora_ML, Cite-seer, Pubmed, and ACM are citation datasets and belong to attributed graphs. BlogCatalog and Polblogs are social network graphs, where the former is an attributed graph and the latter is an unattributed graph. The basic statistics of these datasets are provided in Table 2.

Table 2: The basic statistics of graph datasets

Dataset	Nodes	Edges	Classes	Features
Cora	2,708	5,429	7	1,433
Cora_ML	2,995	16,316	7	2,879
Polblogs	1,490	19,025	2	/
Citeseer	3,327	4,732	6	3,703
Pubmed	19,717	44,338	3	500
ACM	3,025	13,128	3	1,870
BlogCatalog	5,196	171,743	6	8,189

Implementation details. All of the baseline methods come from the code library PyG: https://github.com/pyg-team/pytorch_geometric and DeepRobust: <https://github.com/DSE-MSU/DeepRobust>. All experiments run on a server equipped with an Intel® Xeon® Gold 6254 CPU @ 3.10GHz and NVIDIA GeForce RTX 3090 GPU.

A.1.2 Baseline Model. To evaluate the effectiveness of the HSIC-GAD, we compare it with the following classical GNNs and defense models, the introduces of model are as follows:

- **GCN[22]:** This model defines graph convolution as two steps: aggregation and transformer. It is the most representative GNN model.
- **GAT[30]:** This model utilizes a multi-head attention mechanism to assign different weights to the neighbors of the node. It is often used as a baseline to defend against adversarial attacks.
- **JKNet[36]:** JKNet maximizes or concatenates the representations of each node in the last layer, and divides them into JKNetMaxpool and JKNetConcat based on the different processing methods of the last layer.
- **RGCN[40]:** RGCN uses Gaussian distribution as the hidden representation of nodes in graph convolution layers and assigns attention weights to neighboring nodes based on their variance.
- **GCN-Jaccard[33]:** Since attackers tend to modify the topology of the graph, they often perturb the model by connecting nodes with different features or labels. Based on the above findings, the model utilizes the Jaccard similarity of node features to purify the graph structure.
- **GCN-SVD[3]:** The author analyze Netattack and find that it is a high-order attack. Based on this analysis, they propose a graph purification strategy that utilizes singular value decomposition of the adjacency matrix, preserving only low-order approximations.
- **Pro-GNN[20]:** A general framework Pro-GNN can learn a robust graph neural network model from perturbed graph guided by constraining graph data to be low-rank, sparse, and feature smoothing.
- **Elastic[25]:** The author proposes an elastic message passing scheme and integrates it with deep neural networks leading to elastic GNNs that are significantly robust to graph adversarial attacks.

- **Mid-GCN[17]:** Mid-GCN discovers the importance of mid-frequency signals on the robustness of GNNs. It is a state-of-the-art defense GNN in node classification accuracy under various adversarial attack strategies.

A.1.3 Adversarial Attack Methods. We compare the robustness of HSIC-GAD not only against IMAs, but also under other types of adversarial attacks. The detail of attack strategies are as follows:

- **Netattack[41]:** This method is a classical approach in hybrid attacks, generating adversarial samples based on a surrogate model. It is designed for targeted attacks and can perform both evasion and poison attacks. To maintain fairness, we adopt the target node selection strategy from Netattack in all target attack methods.
- **IG-FGSM[33]:** This method generates adversarial attacks based on integrated gradients. They believe that integrated gradients can better reflect the effect of perturbing certain features or edges.
- **Metattack[42]:** The author proposes a novel poisoning attack that significantly degrades global node classification performance. This attack is formulated as a quadratic optimization problem and utilizes meta-gradients to identify the optimal perturbations that lead to a marked decrease in model performance.
- **RWCS, GC-RWCS[27]:** This method establishes a connection between adversarial attacks and random walks and uses greedy strategies to search for perturbed node sets based on random walks.
- **InfMax-Norm, InfMax-Unif[26]:** This method considers adversarial attacks on GNNs as an influence maximization problem and uses the theory of influence maximization problem to analyze adversarial attacks, assigning the optimal set of perturbation nodes under the guidance of the idea of influence maximization.
- **VoteAttack[32]:** Under the guidance of influence maximization theory, this method improves the voting algorithm and selects a set of influential nodes that can achieve optimal perturbation performance. It is currently the most effective model in influence maximization attacks.

A.1.4 Evaluating Metrics.

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(f_{\theta}(\mathbf{A}, \mathbf{X})_i = \mathbf{Y}_i)}{N}, \quad (11)$$

where $\mathbb{I}(\cdot)$ is the indicator function, N is the number of nodes in the graph, $f_{\theta}(\mathbf{A}, \mathbf{X})_i$ is the prediction of the model on the i -th node. Higher model accuracy indicates stronger adversarial robustness under adversarial attacks.

A.2 Additional Results

A.2.1 Ablation Experiment. The effectiveness of the framework is independent of the choice of backbone models. To evaluate its generalizability, we conduct experiments on various backbones using different datasets. Table 3 shows the results, where models marked with an asterisk (*) indicate integration with HSIC-GAD. All backbones perform better when combined with the framework, demonstrating its strong generalizability.

Table 3: Robustness of HSIC-GAD across different backbone models

Dataset	Method Model	GCN	GCN*	Jaccard	Jaccard*	SVD	SVD*	Elastic	Elastic*	Mid- GCN	Mid- GCN*
Cora	None	86.32	86.69	80.41	90.76	82.07	82.62	85.40	87.62	83.36	83.36
	Degree	71.72	86.69	80.40	90.76	79.48	80.78	85.03	87.25	83.18	82.99
	PageRank	63.96	79.67	78.37	86.14	54.52	66.17	78.93	80.04	75.05	77.82
	Between	67.84	80.22	78.00	87.43	59.88	69.69	78.93	80.59	74.86	76.16
	Random	77.26	80.04	76.52	86.69	56.19	67.84	77.26	78.56	74.49	77.08
	GC-RWCS	58.96	79.48	77.82	86.69	57.85	66.91	79.30	80.96	74.68	77.08
	RWCS	63.96	79.85	78.19	86.14	55.63	66.17	79.30	80.04	74.49	77.26
	InfMax-Unif	59.89	80.04	77.82	87.06	52.12	62.66	76.71	78.56	73.94	75.42
	InfMax-Norm	59.52	80.22	78.74	88.35	55.45	64.69	78.56	80.41	74.68	76.71
	VoteAttack	57.30	80.60	77.82	85.58	52.86	64.69	78.74	80.22	74.86	76.71
Citeseer	None	76.39	78.50	74.14	83.31	72.93	73.38	75.19	76.54	70.53	70.53
	Degree	69.47	78.35	72.33	83.31	47.37	57.59	72.93	75.49	64.51	64.21
	PageRank	54.74	71.13	67.52	79.84	39.70	46.77	57.29	61.80	53.83	56.54
	Between	58.05	72.18	69.92	81.20	43.61	48.87	60.90	66.32	55.64	56.69
	Random	70.08	72.18	71.42	80.45	55.94	61.05	69.62	72.78	63.76	64.81
	GC-RWCS	55.34	74.44	68.12	78.80	37.89	44.36	58.35	60.90	52.48	56.99
	RWCS	54.29	71.13	67.67	80.00	40.00	47.07	57.89	61.80	54.13	56.54
	InfMax-Unif	58.80	72.48	69.02	79.85	40.30	46.77	58.35	63.16	55.04	56.99
	InfMax-Norm	56.54	72.48	68.12	78.80	41.65	45.86	57.44	60.75	52.48	56.09
	VoteAttack	54.44	71.43	66.31	78.95	37.44	45.56	55.19	60.60	51.88	55.49
Pubmed	None	86.33	86.03	87.01	87.29	84.50	84.53	84.78	83.34	84.18	82.59
	Degree	70.38	86.05	69.54	71.77	48.49	43.04	62.14	70.20	54.98	59.08
	PageRank	64.22	84.71	63.27	69.14	48.64	49.35	64.82	73.55	48.63	71.18
	Between	64.80	84.88	62.08	66.29	47.48	48.34	65.28	75.91	49.73	72.70
	Random	75.04	84.76	76.61	79.74	64.01	70.48	81.53	80.40	63.65	78.00
	GC-RWCS	63.07	84.96	61.04	62.67	47.07	48.82	66.14	77.25	47.77	72.10
	RWCS	63.00	84.66	62.03	67.11	47.48	48.41	63.41	73.78	48.42	71.39
	InfMax-Unif	63.35	85.04	60.25	63.28	47.35	49.78	69.62	77.12	47.00	73.32
	InfMax-Norm	62.26	84.99	59.90	61.78	47.65	49.76	67.06	77.02	46.07	73.54
	VoteAttack	65.89	84.96	61.14	62.14	49.04	50.85	65.00	72.91	47.95	67.15

Table 4: Performance and runtime comparison across different datasets

		GCN	SVD	RGCN	Elastic	Mid-GCN	HSIC-GAD
Cora	accuracy(%)	57.30	52.86	82.25	82.80	76.89	85.95
	time(s)	0.58	0.60	1.48	12.13	3.22	3.24
Citeseer	accuracy(%)	54.44	37.44	73.08	55.19	64.66	78.35
	time(s)	0.59	0.59	1.51	12.38	3.24	3.44
Pubmed	accuracy(%)	65.89	49.04	76.29	65.00	47.95	79.93
	time(s)	8.56	7.17	13.95	14.28	5.91	20.74

A.2.2 Adversarial Robustness under Topology Attacks. We further evaluate the adversarial robustness of HSIC-GAD against topology attacks. Note that, consistent with prior works [8, 20, 41, 42], all experiments on topology attacks are performed on the largest connected component (LCC) of the graphs. To demonstrate that HSIC-GAD is resistant to different levels of topology perturbation, we vary the perturbation rate as {5%, 10%, 15%, 20%} for topology attacks, and compare its robustness with that of baseline methods. For each experiment, we report the average performance and standard error over 10 runs. The experimental results in Figure 7 show that the model performance remains high under different

structural perturbation rates, and as the perturbation rate increases, the performance of HSIC-GAD remains stable compared to other models. This demonstrates that HSIC-GAD achieves competitive performance against topology attacks.

A.2.3 Adversarial Robustness under Extreme Budget Scenarios. We further investigate the defensive performance of HSIC-GAD under extreme attack scenarios, where the number of perturbed nodes was increased tenfold. For the adversary, we employed VoteAttack [32], a state-of-the-art IMA. As shown in the Table 4, even under such

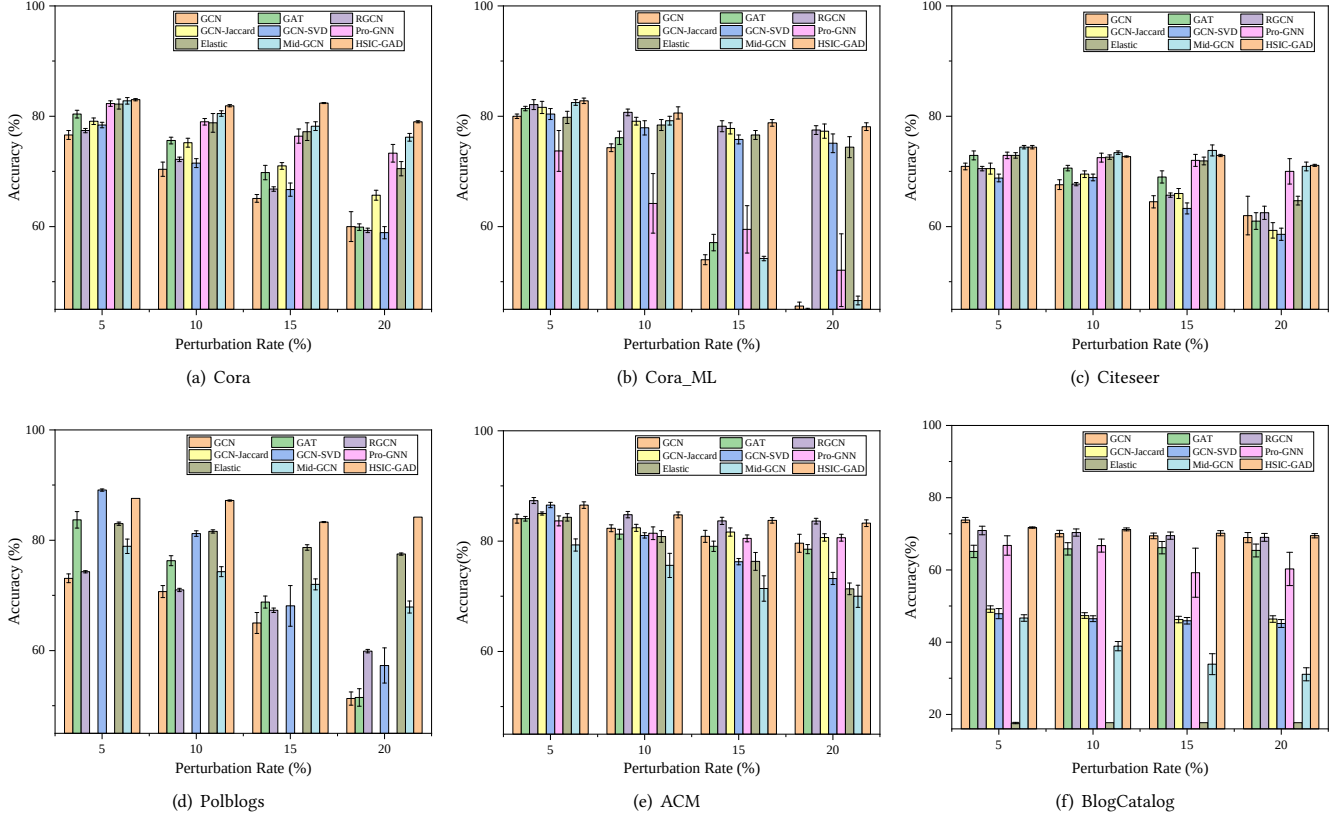


Figure 7: Robustness of various models under different edge perturbation rates.

Table 5: Performance under extreme perturbation budgets across different datasets

Model \ Dataset	GCN	GAT	JKNet	Jaccard	SVD	RGCN	Elastic	Mid-GCN	Ours
Cora	33.46	<u>60.26</u>	39.23	39.00	35.90	27.36	27.73	35.12	77.26
Citeseer	24.06	33.25	32.23	32.03	<u>36.86</u>	36.69	29.77	36.54	61.50
Pubmed	40.22	34.09	40.73	38.80	<u>43.58</u>	42.84	38.60	40.21	74.81

extreme perturbation budgets, the proposed method consistently maintains strong defensive performance.

A.2.4 Computational Complexity. This subsection investigates the runtime efficiency of the proposed method across different datasets. The computational complexity of HSIC-GAD originates from the calculation of the kernel matrix. To reduce its computational complexity, this paper employs a mini-batch and random neighbor sampling strategy. Specifically, the minibatch size is 512, and each

node samples only a small and fixed number of neighboring nodes, which effectively avoids the risk of memory overflow. In the computation of the kernel matrix, we also adopt the size of the batch for the calculation, which ensures the time efficiency of HSIC-GAD. The accuracy of each model under VoteAttack and the running time are detailed in Table 5. The results indicate that the proposed HSIC-GAD exhibits strong adversarial robustness while maintaining a controllable runtime.